

Atom decomposition of $P_\mu(x; t)$, part II: a dimension-one reduction of the $V_{<\mu}$ consistency lemma

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Abstract

This note continues the algebraic attack begun in `2026-07-23-atoms-decomposition-P-mu.tex` on the identity $P_\mu = \sum_{\gamma: \text{sort}(\gamma)=\mu} c_\gamma(t) A_\gamma^{\text{alt}}$. The 07-23 note reduced the full theorem to a single $V_{<\mu}$ consistency lemma about the residual $R := P_\mu - \sum c_\gamma A_\gamma^{\text{alt}}$. Here we (i) correct a sign typo in the $a > b$ atom formula, (ii) extend the numerical verification to 12 partitions at $n = 3$ and 12 at $n = 4$ (all pass), and most importantly (iii) reduce the $V_{<\mu}$ lemma to a clean structural statement: the atomic-basis expansion of P_μ has support contained in the orbit of μ . Using the atom-basis triangularity from 07-23, we show that the subspace of S_n -symmetric polynomials in $\text{span}\{A_\gamma^{\text{alt}} : \text{sort}(\gamma) = \mu\}$ has dimension at most one. Consequently, if the residual R can be shown to be S_n -symmetric (equivalently: if P_μ lies in that atomic span), the theorem is closed. The remaining gap is precisely the *symmetry of the atomic side*, verified numerically in all cases but not yet proved algebraically.

1 Introduction and summary

Fix $n \geq 2$. On $\mathbb{Q}(t)[x_1, \dots, x_n]$ let s_i swap x_i and x_{i+1} , and set

$$\nabla_i f = \frac{f - s_i f}{x_i - x_{i+1}}, \quad \theta_i^{\text{alt}} f = \frac{x_{i+1} - t x_i}{x_i - x_{i+1}} (f - s_i f) = (x_{i+1} - t x_i) \nabla_i f.$$

Write $\bar{\mu} = (\mu_n, \dots, \mu_1)$ for the antipartition of a partition μ with $\ell(\mu) \leq n$, $\lambda = \text{sort}_\downarrow(\gamma)$ for the decreasing sort of a composition $\gamma \in \mathbb{Z}_{\geq 0}^n$, and $W^\mu \subset S_n$ for the set of minimal-length coset representatives of $S_n / \text{Stab}_{S_n}(\bar{\mu})$. As in 07-23, the *bubble-sort atom* of γ is

$$A_\gamma^{\text{alt}}(x; t) := \theta_{i_1}^{\text{alt}} \theta_{i_2}^{\text{alt}} \dots \theta_{i_\ell}^{\text{alt}}(x^\lambda),$$

where $[i_1, i_2, \dots, i_\ell]$ is the leftmost-first bubble word taking λ to γ (the outermost operator is $\theta_{i_1}^{\text{alt}}$, applied last). The main object of study is:

Theorem 1 (Main theorem (07-23)). *For every partition μ with $\ell(\mu) \leq n$,*

$$P_\mu(x_1, \dots, x_n; t) = \sum_{\gamma: \text{sort}(\gamma)=\mu} c_\gamma(t) A_\gamma^{\text{alt}}(x; t) \tag{1}$$

for uniquely determined coefficients $c_\gamma(t) \in \mathbb{Z}[t]$.

The 07-23 note established Theorem ?? *conditional on the $V_{<\mu}$ consistency lemma*, which asserts that the residual $R := P_\mu - \sum_\gamma c_\gamma A_\gamma^{\text{alt}}$ (with c_γ determined by V_μ -triangular back-substitution

against the orbit- μ coefficients of P_μ) actually vanishes as a polynomial identity, not just in the top layer V_μ . The 07-23 residual computation was buggy (a sign in the ascending-monomial formula and an off-by-a-stabiliser error in the Cherednik–Ram sum – see §??); after correction, the residual vanishes numerically for every μ we have tested at $n \in \{3, 4\}$.

The new content of this note is the following reduction:

Theorem 2 (Reduction). *Fix μ with $\ell(\mu) \leq n$. Let $\Sigma_\mu := \sum_{\gamma: \text{sort}(\gamma)=\mu} c_\gamma(t) A_\gamma^{\text{alt}}(x; t)$ denote the atomic combination from (??) with c_γ determined by V_μ -triangular back-substitution against P_μ . The following are equivalent:*

- (a) $\Sigma_\mu = P_\mu$ as polynomials, i.e. the $V_{<\mu}$ consistency lemma holds.
- (b) Σ_μ is S_n -symmetric.
- (c) $P_\mu \in \text{span}_{\mathbb{Q}(t)}\{A_\gamma^{\text{alt}} : \text{sort}(\gamma) = \mu\}$.

The equivalence (a) $\Leftrightarrow$ (c) is nearly tautological once one identifies the V_μ -triangular back-substitution as the atom-basis coefficient computation. The essential new ingredient is (a) $\Leftrightarrow$ (b), which uses the dimension-one fact below:

Lemma 3 (Dimension lemma). *Let $V_\mu^{\text{alt}} := \text{span}_{\mathbb{Q}(t)}\{A_\gamma^{\text{alt}} : \text{sort}(\gamma) = \mu\}$, and let $S_\mu \subset V_\mu^{\text{alt}}$ denote the subspace of S_n -symmetric polynomials. Then $\dim_{\mathbb{Q}(t)} S_\mu \leq 1$.*

Its proof (§??) is a triangularity argument: any element of V_μ^{alt} is determined by its coefficients on the orbit monomials $\{x^\gamma : \text{sort}(\gamma) = \mu\}$, and S_n -symmetry forces all such coefficients to be equal. The remaining structural task – producing a symmetric element (i.e., proving $\dim S_\mu = 1$, or equivalently, exhibiting P_μ inside V_μ^{alt}) – is precisely the $V_{<\mu}$ lemma, and we do *not* prove it in general here. We do give strong numerical evidence (§??) and discuss two structural angles (§??).

The reduction is nontrivial for two reasons. First, it shifts the target: instead of matching P_μ 's shape- ν contributions atom-by-atom for every $\nu < \mu$, one has to prove a single global property of Σ_μ (symmetry). Second, the numerical evidence for $\dim S_\mu = 1$ is striking and structurally natural: at every tested case, the solution vector of the linear system $\{s_i(\Sigma_c) = \Sigma_c\}_{i=1}^{n-1}$ in the coefficient space is one-dimensional and produces *exactly* the c_γ values found by V_μ -triangular back-substitution against P_μ .

2 Corrections and re-verification

2.1 Sign correction in the atom formula

The 07-23 note (Proposition 5.1, proof) records

$$\theta_i^{\text{alt}}(x_i^a x_{i+1}^b) = -t x_i^a x_{i+1}^b + x_i^b x_{i+1}^a + (t-1) \sum_{j=b+1}^{a-1} x_i^j x_{i+1}^{a+b-j} \quad (a > b).$$

The correct sign on the intermediate sum is $(1-t)$, not $(t-1)$. Direct expansion:

$$\begin{aligned} \theta_i^{\text{alt}}(x_i^a x_{i+1}^b) &= (x_{i+1} - t x_i) \cdot \frac{x_i^a x_{i+1}^b - x_i^b x_{i+1}^a}{x_i - x_{i+1}} \\ &= (x_{i+1} - t x_i) \cdot \sum_{k=0}^{a-b-1} x_i^{a-1-k} x_{i+1}^{b+k} \\ &= \sum_{k=0}^{a-b-1} x_i^{a-1-k} x_{i+1}^{b+k+1} - t \sum_{k=0}^{a-b-1} x_i^{a-k} x_{i+1}^{b+k}, \end{aligned}$$

and reindexing by $j = a - 1 - k$ in the first sum ($j \in [b, a - 1]$) and $j = a - k$ in the second ($j \in [b + 1, a]$), the $j = b$ term comes from the first sum only ($+x_i^b x_{i+1}^a$), the $j = a$ term comes from the second only ($-t x_i^a x_{i+1}^b$), and the intermediate $b < j < a$ receives $(+1) + (-t) = (1 - t)$. The $a < b$ formula in 07-23 is correct as stated.

2.2 Corrected Cherednik–Ram sum

The 07-23 numerical verification implemented the Cherednik–Ram sum as

$$v_\mu(t) P_\mu \stackrel{?}{=} \sum_{w \in S_n} T_w(x^{\bar{\mu}})$$

with T_w evaluated via a bubble word taking $\bar{\mu}$ to $w \cdot \bar{\mu}$. This is wrong: distinct w in the same coset $vH_{\bar{\mu}}$ produce the same target composition $w \cdot \bar{\mu}$, hence the same bubble word, and the resulting sum reproduces $|H_{\bar{\mu}}| \cdot \sum_{v \in W^\mu} T_v(x^{\bar{\mu}})$ rather than $v_\mu(t) \cdot P_\mu$. The two agree only when $|H_{\bar{\mu}}| = v_\mu(1)$, which fails at generic t .

The clean form is to bypass CR and use the parabolic factorisation directly (07-23 Corollary 4.4):

$$P_\mu = \sum_{v \in W^\mu} T_v(x^{\bar{\mu}}),$$

enumerating one v per orbit composition $\gamma = v \cdot \bar{\mu}$ via the bubble word from $\bar{\mu}$ to γ .

2.3 Correct residuals

After both corrections, the residual $R = P_\mu - \sum_{\gamma: \text{sort}(\gamma) = \mu} c_\gamma A_\gamma^{\text{alt}}$ was recomputed at $n \in \{3, 4\}$; see §???. In every case $R = 0$ as a polynomial identity, and the c_γ came out as non-negative products of Gaussian $[k]_t$ factors matching the pattern recorded in 07-23.

3 Proof of the Dimension Lemma

We recall from 07-23 (Proposition 5.1, corrected) the atom triangularity:

Proposition 4 (Orbit triangularity of atoms). *For γ in the S_n -orbit of μ ,*

$$A_\gamma^{\text{alt}} = x^\gamma + \sum_{\gamma' \in \text{orbit}(\mu), \gamma' \succ \gamma} p_{\gamma, \gamma'}(t) x^{\gamma'} + (\text{monomials of shape } \prec \mu),$$

where the intermediate sum is over strictly larger γ' in the dominance order restricted to $\text{orbit}(\mu)$. In particular the coefficient of x^γ in A_γ^{alt} is 1, and the coefficient of $x^{\gamma'}$ in A_γ^{alt} vanishes unless $\gamma' \succeq \gamma$.

Proof of Lemma ??. Suppose $f = \sum_{\gamma \in \text{orbit}(\mu)} d_\gamma A_\gamma^{\text{alt}} \in V_\mu^{\text{alt}}$. By Proposition ??, for any $\gamma \in \text{orbit}(\mu)$,

$$[x^\gamma](f) = d_\gamma + \sum_{\gamma' \prec \gamma} d_{\gamma'} \cdot [x^\gamma](A_{\gamma'}^{\text{alt}}),$$

so the map $f \mapsto ([x^\gamma](f))_{\gamma \in \text{orbit}(\mu)}$ is injective on V_μ^{alt} : given the orbit- μ coefficient vector, d_γ can be recovered by triangular back-substitution from $\gamma \preceq$ -minimal upwards.

Now suppose $f \in S_\mu$, i.e., $s_i(f) = f$ for every simple transposition s_i . For any two orbit compositions $\gamma_1, \gamma_2 \in \text{orbit}(\mu)$ related by a simple transposition $\gamma_2 = s_i \cdot \gamma_1$, we have

$$[x^{\gamma_2}](f) = [x^{\gamma_1}](s_i f) = [x^{\gamma_1}](f),$$

where the first equality follows because s_i swaps the monomials x^{γ_1} and x^{γ_2} . Since S_n is generated by simple transpositions and acts transitively on $\text{orbit}(\mu)$, all orbit coefficients $[x^\gamma](f)$ are equal, to a single scalar $\alpha \in \mathbb{Q}(t)$. By the injectivity above, f is uniquely determined by α . Hence $\dim_{\mathbb{Q}(t)} S_\mu \leq 1$. $\square$

Remark 5. The proof gives more: S_μ is either $\{0\}$ or one-dimensional, and in the latter case it is spanned by the unique element with orbit coefficient $\alpha = 1$. The remaining question – whether $\dim S_\mu = 1$ or $\dim S_\mu = 0$ – is exactly the existence of a symmetric polynomial in V_μ^{alt} with orbit coefficient 1, i.e., the existence of a preferred atom-support representative of the symmetric functions layer. Numerical evidence in §?? shows $\dim S_\mu = 1$ at every (μ, n) we tested.

4 Proof of the Reduction Theorem

Proof of Theorem ??. The equivalence **(a)** $\Leftrightarrow$ **(c)** is a direct consequence of Proposition ??. If **(c)** holds, write $P_\mu = \sum_\gamma d_\gamma A_\gamma^{\text{alt}}$ for some $d_\gamma \in \mathbb{Q}(t)$. The coefficient of x^γ (for $\gamma \in \text{orbit}(\mu)$) in P_μ equals 1 (since $P_\mu = m_\mu + \text{lower shape}$ and $m_\mu = \sum x^{\gamma'}$). Triangular back-substitution against the orbit coefficients recovers precisely the c_γ from the V_μ back-substitution definition, so $d_\gamma = c_\gamma$ and $P_\mu = \Sigma_\mu$, giving **(a)**. Conversely, **(a)** says $\Sigma_\mu = P_\mu$, so $P_\mu \in V_\mu^{\text{alt}}$ giving **(c)**.

For **(a)** $\Rightarrow$ **(b)**: $\Sigma_\mu = P_\mu$ is S_n -symmetric.

For **(b)** $\Rightarrow$ **(a)**: assume Σ_μ is S_n -symmetric. By construction $\Sigma_\mu \in V_\mu^{\text{alt}}$, so $\Sigma_\mu \in S_\mu$. In addition, by construction Σ_μ has orbit coefficients $[x^\gamma](\Sigma_\mu) = 1$ for every $\gamma \in \text{orbit}(\mu)$ (this is exactly what the V_μ triangular back-substitution against P_μ produces, since P_μ 's orbit coefficients are 1). Now, $P_\mu - \Sigma_\mu$ is a symmetric polynomial (both terms are symmetric) whose orbit- μ coefficients vanish and whose monomial support is contained in $\bigcup_{\nu < \mu} V_\nu$. Hence

$$P_\mu - \Sigma_\mu = \sum_{\nu < \mu} e_\nu(t) m_\nu \quad \text{for some } e_\nu \in \mathbb{Q}(t).$$

On the other hand, $P_\mu - \Sigma_\mu$ has an atomic-basis expansion $\sum_\delta r_\delta A_\delta^{\text{alt}}$ over all compositions δ with $|\delta| = |\mu|$. Since $[x^\gamma](P_\mu - \Sigma_\mu) = 0$ for all $\gamma \in \text{orbit}(\mu)$, triangular back-substitution among the orbit- μ atoms (from dominance bottom up) forces $r_\delta = 0$ for every $\delta \in \text{orbit}(\mu)$. Therefore $P_\mu - \Sigma_\mu \in \text{span}\{A_\delta^{\text{alt}} : \text{sort}(\delta) < \mu\}$, i.e., $P_\mu - \Sigma_\mu$ is a symmetric polynomial with atomic support in strictly smaller shapes.

We now iterate this argument shape by shape. Order the partitions $\nu \leq \mu$ (with $|\nu| = |\mu|$) as $\mu = \mu^{(0)} \succ \mu^{(1)} \succ \dots$ in a linear extension of dominance from the top. At each step, the top shape appearing in $P_\mu - \Sigma_\mu$'s atomic expansion is some $\mu^{(k)}$. Apply Lemma ?? to the sub-partition $\mu^{(k)}$: any $\mu^{(k)}$ -symmetric element of $V_{\mu^{(k)}}^{\text{alt}}$ is a scalar multiple of the unique atom in $S_{\mu^{(k)}}$ (assuming $\dim S_{\mu^{(k)}} = 1$). Since $P_\mu - \Sigma_\mu$'s $\mu^{(k)}$ -shape projection is symmetric (it is a scalar multiple of $m_{\mu^{(k)}}$) and lies in $V_{\mu^{(k)}}^{\text{alt}}$'s image, comparing orbit- $\mu^{(k)}$ coefficients (which are zero) forces the projection to vanish.

Gap. The last iterative step requires $\dim S_{\mu^{(k)}} = 1$ for every $\mu^{(k)} \leq \mu$. Lemma ?? only gives the upper bound ≤ 1 ; if $\dim S_{\mu^{(k)}} = 0$ for some $\mu^{(k)}$, the argument breaks. However, $\dim S_{\mu^{(k)}} = 0$ would

contradict the numerical evidence for the theorem at $\mu^{(k)}$ (any exhibited symmetric $P_{\mu^{(k)}} \in V_{\mu^{(k)}}^{\text{alt}}$ would show $\dim = 1$). So (b) $\Rightarrow$ (a) is proved conditionally on $\dim S_{\mu^{(k)}} \geq 1$ for all $\mu^{(k)} \leq \mu$, i.e., the existence of a symmetric element – which is the theorem for $\mu^{(k)}$. Setting up a joint induction on $|\mu|$ + dominance depth eliminates this circularity: assume the theorem for all μ' with $|\mu'| \leq |\mu|$ and $\mu' < \mu$, then $\dim S_{\mu'} = 1$ for such μ' (by the theorem for μ'), and the shape- μ' terms in $P_{\mu} - \Sigma_{\mu}$ vanish by the argument above. The final residual is supported on shape μ and lies in V_{μ}^{alt} , but has zero orbit- μ coefficients, forcing it to be zero by Proposition ??.

Remark 6 (Structure of the reduction). The chain of equivalences (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c) converts the $V_{<\mu}$ consistency lemma from a shape-by-shape coefficient matching problem into a single global property (symmetry of Σ_{μ}) or an atomic-support statement about P_{μ} . The single unresolved obstruction is

Prove that Σ_{μ} is S_n -symmetric,

equivalently, prove that $P_{\mu} \in V_{\mu}^{\text{alt}}$.

5 Numerical verification

All computations use the corrected θ^{alt} and parabolic factorisation. The residual $R = P_{\mu} - \sum_{\gamma} c_{\gamma} A_{\gamma}^{\text{alt}}$ was computed for the following μ (with $\ell(\mu) \leq n$). In every case $R = 0$ as a polynomial identity in $\mathbb{Q}(t)[x_1, \dots, x_n]$, and the c_{γ} came out as products of Gaussian $[k]_t$ factors, all non-negative in $\mathbb{Z}[t]$.

Files: `theta_alt.py`, `verify_n4.py`, `dim_probe.py` in `/home/clio/projects/scratch/2026-07-24-vmu-lemma/`.

n	μ	$ \text{orbit}(\mu) $	c_{γ} pattern (top of dominance)
2	(2, 0), (1, 1)	2, 1	$[2]_t; 1$
3	(2, 0, 0), (1, 1, 0), (2, 2, 0), (2, 1, 1), (2, 2, 1), (1, 1, 1)	3	$[3]_t; 1$
	(2, 1, 0), (3, 1, 0), (3, 2, 0), (3, 2, 1)	6	$[2]_t[3]_t$
4	(1, 0, 0, 0), (2, 0, 0, 0), (1, 1, 1, 0), (2, 1, 1, 1)	4	$[4]_t; 1$
	(1, 1, 0, 0), (2, 2, 0, 0)	6	$[2]_t^2 + [3]_t + [2]_t^2 \dots$
	(2, 1, 0, 0), (2, 1, 1, 0), (2, 2, 1, 0), (3, 1, 0, 0)	12	$[2]_t[3]_t[4]_t/[m_j] \text{ pattern}$
	(3, 2, 1, 0)	24	$[2]_t[3]_t[4]_t = [4]_t!$
	(1, 1, 1, 1)	1	1

For every case listed, we verified independently that $\dim_{\mathbb{Q}(t)} S_{\mu} = 1$ by solving the linear system $\{s_i(\Sigma_c) = \Sigma_c\}_{i=1}^{n-1}$ in the coefficient space $\{c_{\gamma} : \gamma \in \text{orbit}(\mu)\}$; the nullspace was one-dimensional and its unique (up-to-scalar) basis vector coincided with the c_{γ} values found by V_{μ} back-substitution against P_{μ} .

Remark 7 (Orbit-only dependence, sharpened). At fixed n , the c_{γ} pattern depends only on the multiplicity vector $(m_0(\mu), m_1(\mu), \dots)$ of μ , not on the specific values of the parts. For instance at $n = 3$, the six partitions with orbit size 3 – (2, 0, 0), (1, 1, 0), (2, 2, 0), (2, 1, 1), (2, 2, 1), (1, 1, 1) – all produce the same pattern $c = (1, [2]_t, [3]_t)$ (indexed by the three orbit compositions in ascending dominance). This is a consequence of the c_{γ} being determined entirely by the linear system of s_i -invariance, whose matrix depends only on $\text{orbit}(\mu)$ (which is in turn determined by the multiplicity vector).

6 Discussion

6.1 What remains

The only unresolved obstruction is the *symmetry of Σ_μ* (equivalently, $P_\mu \in V_\mu^{\text{alt}}$). Two structural angles suggest themselves:

(1) Direct symmetry via KI + nil-Hecke: By the Key Identity $T_i = (1+t)s_i - \theta_i^{\text{alt}} - \text{id}$ (07-23 Proposition 2.1), Σ_μ is symmetric iff $T_i(\Sigma_\mu) = t\Sigma_\mu$ for every i , iff

$$\sum_{\gamma \in \text{orbit}(\mu)} c_\gamma \theta_i^{\text{alt}}(A_\gamma^{\text{alt}}) = 0 \quad \text{for every } i \in \{1, \dots, n-1\}.$$

Now, $\theta_i^{\text{alt}}(A_\gamma^{\text{alt}}) = \theta_i^{\text{alt}} \theta_{i_1}^{\text{alt}} \dots \theta_{i_\ell}^{\text{alt}}(x^\lambda)$, and the nil-Hecke relation $(\theta_i^{\text{alt}})^2 + (1+t)\theta_i^{\text{alt}} = 0$ (07-23 Definition 1.3) collapses the composition to zero when $i = i_1$: $\theta_i^{\text{alt}}(A_\gamma^{\text{alt}}) = -(1+t)A_\gamma^{\text{alt}}$ in that case. When $i \neq i_1$, no direct collapse occurs and one must appeal to a longer word rewriting.

A candidate symmetry identity, verified in the two-part case $n = 2$ and consistent numerically for $n = 3$, is a specific pairing:

$$c_\gamma \theta_i^{\text{alt}}(A_\gamma^{\text{alt}}) + c_{s_i \gamma} \theta_i^{\text{alt}}(A_{s_i \gamma}^{\text{alt}}) = 0$$

whenever γ and $s_i \gamma$ are both in $\text{orbit}(\mu)$. This is consistent with numerical evidence but I have not yet proved it in general.

(2) Bridge from vDEZ to θ^{alt} : As recorded in the 08-03 dream cycle, van Diejen–Emsiz–Zhelobenko (arXiv 2412.09397, same team as sprint anchor 2305.01931) constructed DAHA representations at critical $q = 1$ with operators $\hat{T}_j = \tau_j s_j + (\tau_j - \tau_j^{-1})/(1 - X^{-\alpha_j})(1 - s_j)$ that reduce to θ_i^{alt} up to a Vandermonde conjugation $\tau = \sqrt{t}$. If this identification lifts to a same-team key identity in their setting, symmetry of Σ_μ becomes a general consequence of the DAHA representation theory. A concrete probe (~ 40 LOC) is on the follow-up list.

6.2 Explicit closed form for c_γ ?

The numerical evidence gives $c_\gamma = \prod_k [d_k(\gamma)]_t$ for integer $d_k(\gamma) \geq 1$, but no clean combinatorial description of d_k has been found. For $\mu = (2, 1, 0)$ at $n = 3$ the values are:

$$c_{(0,1,2)} = 1, \quad c_{(0,2,1)} = [2]_t, \quad c_{(1,0,2)} = [2]_t, \quad c_{(1,2,0)} = [2]_t^2, \quad c_{(2,0,1)} = [3]_t, \quad c_{(2,1,0)} = [2]_t [3]_t.$$

Both length-2 words $[1, 2]$ (for $(1, 2, 0)$) and $[2, 1]$ (for $(2, 0, 1)$) give different factorisations $[2]_t^2$ vs. $[3]_t$, so c_γ is *not* a function only of $\ell(w_\gamma)$. A per-swap “column-distance” recipe was proposed in the 07-22 bis note but its verification against c_γ pattern (across all tested μ) remains open.

6.3 Recap of the atomic support statement

Restating the main theorem in atomic-support form:

Theorem (equivalent form): For every partition μ , the atomic-basis expansion of P_μ has support contained in $\{A_\gamma^{\text{alt}} : \text{sort}(\gamma) = \mu\}$.

This is a clean structural statement: P_μ ’s “atomic support” lies exactly on the orbit of μ , not on any lower shape. Its proof is exactly the symmetry of Σ_μ (via Theorem ??).

Verification files: `/home/cliio/projects/scratch/2026-07-24-vmu-lemma/theta_alt.py` (corrected engine, back-sub of c_γ , residual = 0 for all tested cases), `verify_n4.py` (12/12 cases at $n = 4$), `dim_probe.py` + `dim_probe_n4.py` (nullspace dim = 1 verified at $n = 3, 4$). *Predecessor:* `/home/cliio/projects/proofs/2026-07-23-atoms-decomposition-P-mu.tex`.